

Hands-on Lab: Pre-Deployment Evaluation – Amazon Bedrock (Nova Pro vs Nova Lite) on BFSI QnA Dataset

Scenario:

You are an AI QA Engineer at Contoso Bank responsible for validating a customer-facing banking assistant that answers questions related to loans, KYC, fraud prevention, and fund transfers. Before deploying the chatbot to customers, the bank wants to evaluate two candidate models on Amazon Bedrock: Amazon Nova Pro and Amazon Nova Lite. Since Nova Lite is more cost-efficient, your task is to run automated evaluations using Amazon Bedrock's LLM-as-a-Judge capability and verify whether the lower-cost model meets the bank's minimum quality threshold (e.g., Correctness score ≥ 0.85) before approving it for production deployment.

Description:

In this lab, you will perform a pre-deployment evaluation of generative AI models using Amazon Bedrock Model Evaluations. You will upload a BFSI QnA dataset in JSONL format to Amazon S3, configure evaluation metrics including Correctness, Completeness, Coherence, Helpfulness, and Relevance, and execute automated evaluation jobs using an evaluator model (LLM as a judge). After running the evaluation, you will analyze the evaluation scorecards, compare the performance of both models, and verify CloudWatch monitoring and cost management controls.

Contents

1. Introduction & Key Concepts
2. Prerequisites & Model Access
3. Cost to Perform this Lab
4. Create S3 Bucket and Upload BFSI QnA Dataset
5. Configure and Run Model Evaluation in Amazon Bedrock
 - 5.1 Configure Evaluation for Amazon Nova Pro
 - 5.2 Configure Evaluation for Amazon Nova Lite

6. Interpret and Compare Evaluation Results
7. CloudWatch Monitoring & Cost Optimization
8. Clean Up & Delete Resources
9. Summary

1. Introduction & Key Concepts

1.1 Amazon Bedrock

Amazon Bedrock is a fully managed AWS service that provides access to high-performance foundation models (FMs) from leading AI providers (such as Amazon, Anthropic, Meta, and AI21) through a single unified API. It enables organizations to build, evaluate, and scale generative AI applications with built-in security, privacy, and responsible AI guardrails.

1.2 Model Evaluation in Amazon Bedrock

Model evaluation is the process of measuring model quality and accuracy before production deployment:

- **Automatic Evaluation (LLM as a Judge):** Uses a separate, highly reliable evaluator model (such as Amazon Nova Lite or Anthropic Claude) to automatically score generated responses against reference ground truth answers.
- **Core Quality Metrics:**
 - **Correctness:** Evaluates whether the generated response is factually accurate compared to the ground truth.
 - **Completeness:** Assesses if the model fully answered all parts of the user prompt.
 - **Coherence:** Measures logical organization, structure, and readability.
 - **Helpfulness:** Evaluates how useful and actionable the response is for the user.
 - **Relevance:** Assesses how directly and accurately the output addresses the specific input query.

2. Prerequisites & Model Access

1. **AWS Account:** An active AWS account with IAM Administrative access.
2. **Region Selection:** Ensure your AWS Management Console is set to **US East (N. Virginia)** `us-east-1` or another region supporting Bedrock Model Evaluations.
3. **Enable Bedrock Model Access:**
 - In AWS Console, search **Amazon Bedrock**.
 - In left navigation panel, scroll down and click **Model access**.
 - Click **Modify model access** or **Enable specific models**.
 - Select **Amazon Nova Pro**, **Amazon Nova Lite**, and **Anthropic Claude 3 Haiku**.
 - Click **Save changes** and verify status shows **Access granted**.

3. Cost to Perform this Lab

- **Estimated Cost:** ~\$0.01 to \$0.05 (covered by AWS Free Tier or new account credits).
- **Pricing Breakdown:** Pay-per-token pricing for model invocation and evaluation judge inference during evaluation runs.

4. Create S3 Bucket and Upload BFSI QnA Dataset

1. Open AWS Console → Search **S3** → Click **Create bucket**.
2. **Bucket Name:** `contosobfsibedrockeval[your-unique-id]`
3. **AWS Region:** Select **US East (N. Virginia)** `us-east-1`.
4. Leave default settings and click **Create bucket**.
5. Open your newly created bucket and create two subfolders:
 - Click **Create folder** → Name: `eval-input/` → Click **Create folder**.
 - Click **Create folder** → Name: `eval-output/` → Click **Create folder**.
6. **Prepare the Dataset (`bfsi-eval-dataset.jsonl`):**

Create or save a local file `bfsi-eval-dataset.jsonl` with JSON lines formatted for Bedrock Model Evaluation:

```
{  
  "prompt": "Am I eligible for a home loan?",  
  "category": "home loan",  
  "referenceResponse": "Eligibility depends on net salary, credit score, and existing EMI commitments."  
}  
{  
  "prompt": "What documents are required for KYC?",  
  "category": "KYC",  
  "referenceResponse": "KYC requires Government ID proof, Address proof, and recent passport photo."  
}  
{  
  "prompt": "What is the penalty for premature FD withdrawal?",  
  "category": "withdrawal",  
  "referenceResponse": "Premature FD withdrawal incurs a 0.5% to 1.0% interest rate penalty."  
}
```

7. Upload Dataset:

- Open folder `eval-input/` in S3.
- Click **Upload** → Click **Add files** → Select `bfsi-eval-dataset.jsonl` → Click **Upload**.
- Verify path:
`s3://contosobfsibedrockeval[your-unique-id]/eval-input/bfsi-eval-dataset.jsonl`.

5. Configure and Run Model Evaluation in Amazon Bedrock

5.1 Configure Evaluation for Amazon Nova Pro

1. Navigate to **Amazon Bedrock** service in AWS Console.
2. In left navigation panel, under **Assessment**, click **Evaluations**.
3. Select the **Models** tab → Click **Create** → Select **Automatic (model-as-a-judge)**.
4. **Job Configuration:**
 - **Job name:** `bfsi-eval-nova-pro`
5. **Select Evaluator Model (The Judge):**
 - Click **Select model** under Evaluator model.
 - Provider: **Amazon** → Select: **Nova Lite** (or Anthropic Claude 3 Haiku) → Click **Apply**.

6. **Select Generator Model (Model Being Tested):**
 - Click **Select model** under Generator model.
 - Provider: **Amazon** → Select: **Nova Pro** → Click **Apply**.
7. **Select Quality Metrics:**
 - Check the following metrics:
 - Correctness
 - Completeness
 - Coherence
 - Helpfulness
 - Relevance
8. **Configure Input Dataset:**
 - Under **Dataset**, click **Browse S3**.
 - Select bucket `contosobfsibedrockeval[your-unique-id]` → `eval-input` → `bfsi-eval-dataset.jsonl` → Click **Choose**.
9. **Configure Output Location:**
 - Under **Output location**, click **Browse S3**.
 - Select bucket `contosobfsibedrockeval[your-unique-id]` → `eval-output/` → Click **Choose**.
10. **IAM Permissions:**
 - Under Service access, choose **Use an existing service role** or **Create a new service role** (e.g., `BedrockEvaluationServiceRole`).
11. Click **Create**. Job status will show **In progress** (takes ~5–10 minutes).

5.2 Configure Evaluation for Amazon Nova Lite

1. Return to **Evaluations** page → Click **Create** → Select **Automatic (model-as-a-judge)**.
2. **Job Configuration:**
 - **Job name:** `bfsi-eval-nova-lite`
3. **Evaluator Model:** Select **Nova Lite** (matching run 1).
4. **Generator Model:** Select **Amazon Nova Lite**.
5. **Quality Metrics:** Check the following metrics:
 - Correctness
 - Completeness
 - Coherence

- Helpfulness
- Relevance
- 6. **Dataset:** Select
`s3://contosobfsibedrockeval[your-unique-id]/eval-input/bfsi-eval-dataset.jsonl`.
- 7. **Output Location:** Select
`s3://contosobfsibedrockeval[your-unique-id]/eval-output/`.
- 8. **Service Access:** Use the same service role created in step 5.1.
- 9. Click **Create** and wait for job status to show **Completed** (green checkmark).

6. Interpret and Compare Evaluation Results

1. In the **Evaluations** console list, verify both jobs show status **Completed**.
2. Click on `bfsi-eval-nova-pro` to view detailed results:
 - Review **At-a-Glance Scores** (0.0 to 1.0 scale, where 1.0 is perfect).
 - Expected scores for Nova Pro: Correctness (~0.90–0.95), Coherence (~0.95), Helpfulness (~0.90).
3. Click on `bfsi-eval-nova-lite` to view its scorecards:
 - Compare Nova Lite scores against Nova Pro.
 - Check whether Nova Lite meets the bank's minimum **Correctness threshold of 0.85**.
4. **Drill into Prompt Level Details:**
 - Expand specific metrics (e.g., Correctness) and click **Prompt details**.
 - Review input prompts, generated answers, reference responses, and the judge model's natural language reasoning.
5. **Download Detailed JSON Results:**
 - Go to S3 console → `eval-output/` → Open job folder → Download `results.json`.

7. CloudWatch Monitoring & Cost Optimization

1. Set Up CloudWatch Log Group:

- Open **CloudWatch** console → Left panel **Log groups** → Click **Create log group**.
- Log group name: `/aws/bedrock/evaluations`.
- Retention: 7 days.

2. Create CloudWatch Monitoring Dashboard:

- Go to **Dashboards** → Click **Create dashboard** → Name: `Bedrock-BFSI-Eval-Dashboard`.
- Add widgets for Bedrock metrics: `InputTokens`, `OutputTokens`, `ModelInvocations`.

3. Set Up Spending Budget in AWS Budgets:

- Search **AWS Budgets** → Click **Create budget**.
- Budget Name: `Bedrock-Monthly-Budget` → Amount: `$10.00 USD`.
- Add email alert thresholds at 50%, 80%, and 100% of budgeted amount.

8. Clean Up & Delete Resources

1. Delete Evaluation Output Files:

- Go to S3 console → Open `contosobfsibedrockeval[your-unique-id]`.
- Empty and delete the bucket to avoid storage charges.

2. Delete Evaluation Model:

- Go to Amazon Bedrock → Access → Evaluations `Bfsi-eval-nova-lite` and `bfsi-eval-nova-pro`

3. Delete CloudWatch Resources:

- Delete dashboard `Bedrock-BFSI-Eval-Dashboard` and log group `/aws/bedrock/evaluations`.

4. Delete AWS Budget:

- Delete `Bedrock-Monthly-Budget` if no longer needed.

9. Summary

In this hands-on lab, We successfully adapted the pre-deployment generative AI evaluation workflow **Amazon Bedrock** and uploaded a BFSI domain dataset to Amazon S3, configured automated model evaluation jobs using Bedrock's LLM-as-a-Judge capability, evaluated performance metrics (Correctness, Completeness, Coherence, Helpfulness, Relevance), compared scorecards between Amazon Nova Pro and Amazon Nova Lite, and set up AWS CloudWatch monitoring and budget safeguards.